

BOOKLET NUMBER:

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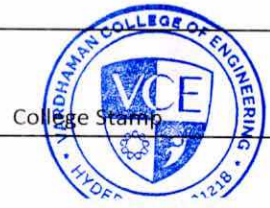


VARDHAMAN

COLLEGE OF ENGINEERING

Shamshabad – 501 218, Hyderabad

CAT EXAMINATION ANSWER BOOKLET

VARDHAMAN COLLEGE OF ENGINEERING
23881A05K0 (C)

A8851 24.10.2025 FN

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CAT - I / II EXAMINATION I / II / III / IV B.Tech / M.Tech / MBA I / II Semester 10/2025
(Strike off whichever is not applicable) (Month and Year)

Name : V. Sathwik

Branch : CCE-c

Subject : Data Science for Engineering

Subject Code : A8851

Date of Exam : 24/10/2025

Signature of the Invigilator with date

(Booklet contains 20 pages)

INSTRUCTIONS TO THE CANDIDATES

- This booklet contains 20 pages. Candidates must ensure it before writing and in case a defective answer book is issued it must be returned to the invigilator and a new and defect free booklet must be obtained.
- Before the candidate begins to answer, registered number, particulars of year, semester, subject, etc., are to be filled in. Failure to enter all or any of these particulars may disqualify the paper from valuation.
- Candidate is prohibited from
 - writing:
 - anything addressing the examiner in any manner whatsoever, in their answer book.
 - objectionable/obscene language in the answer book.
 - anything other than their Registered No. on the question paper.
 - either seeking or providing any assistance to the fellow candidates in the exam.
 - possessing a manuscript or a printed matter, in any form, in the examination hall.
 - bringing loose sheets or paper into the examination hall and detaching any paper from the answer book.
 - carrying Mobile Phone, Smart Watches and any other Electronic Gadgets etc., to Exam Hall.

Violation of these instructions will be viewed as a case of malpractice, which is a punishable offence.

- Before beginning to answer any question, candidates must write the correct question number, and should not write
- Answers must be written legibly on both sides of the paper. There shall be about 25 lines in each page. It is not necessary to begin each answer on a fresh page. Candidates should not use any other ink, except BLACK or BLUE ink.
- Rough work, if any, must be separated, from the subject matter, by a line and noted as rough work.
- The answer book, at the end of the examination, must be handed over to the Assistant Superintendent (Invigilator) by the candidate.

This responsibility lies with the candidate only.

- Candidates should maintain absolute silence during the time of examination. Misbehavior, in any form, by the candidate, in the examination hall, will attract severe punishment.
- Candidates are permitted to leave the examination hall only after the allotted time.
- No additional answer books will be supplied.

To be filled in by the Examiner only

MARKS AWARDED

PART – A				PART – B						
Q No.1				Q No.	2	3	4	5	6	7
a		f		a						
b		g		b						
c		h		c						
d		i		d						
e		j		Total						
Total (a to j)										
In Words:										
In Figures:										

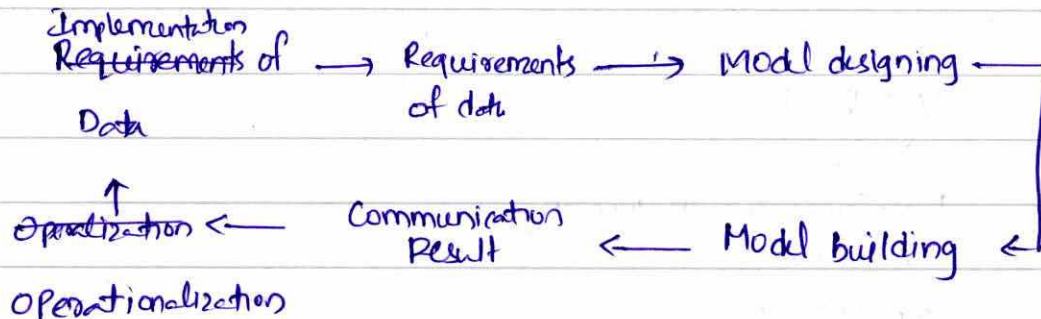
Signature of the Examiner

START WRITING FROM HERE

Q.No.

Part-B

2) Data Analytics life cycle



Implementation of Data

- * It is the first phase of data analytics life cycle
- * All the team members should be aware about the project and implement the meanings
- * Team should implement the model initial model before the data is checked

Requirements of Data

- * It is the second phase of the data analytics life cycle
- * All the team members should know the requirement or techniques
- * This phase is to cleaning the data of testing and training
- * This step is to create the data sets
- * This step will take lot of time to complete.

Q.No.

* This step is done by the human but not ~~not~~ help of machine

Model designing

* It is the third phase of ~~model building~~ data analytics life cycle.

* In this step we ~~are~~ know that how to use the techniques for the given project

* Not only using one technique to implement we use more than one techniques

* The goal is to insights the project

*

Model building :-

* It is the fourth phase of ~~activity~~ life cycle of data analytics

* This step is to test the model

* This process is also called as trail run process

* For the given projects we use tools and techniques for implementing the project (or) Model.

* Tools are

i) MATLAB, ii) Tensorflow, iii) Jupyter, ~~iv)~~

* It should ~~not~~ give both present prediction and information about the feature prediction.

Q.No.

Communication Result:

- * It is the second last phase of data analytics life cycle
- * In this step we should discuss the model to the stakeholders.
- * If the project outcomes is not good. We should communicate clearly with the stakeholder
- * While coming to these step you must have to test the data and train the data.
- * We should explain the model with the ppt presentation. We didn't know that what the expectation should have questioner of stake holder.

Operationalization

- * It is the last step of data analytics life cycle
- * We should deploy the model before the model is tested or trained
- * This is the we should implement projects by us.
- * After deployments the model should work
- * It we should give the updates, etc

Q.No.

3)

Text Analysis Subtask.

They are the different steps in Text Analysis subtask to analyse the meanings of the words and Sentimentation words.

i) Part-of-speech:

It is a category in which they include noun, verb, adjectives etc.

In this step they use to combine the words after the text analysing according to the part of speech.
ex: when the present is noun and verb.

ii) Stemming:

* In this step they use to take root words from the given words or phrases, sentences.

* It is easy way to learn simple.

* Removing the noise words

+

ex: Speaking the root word of speaking is 'speak'.
for working is 'work'.

iii) lemmatization:

In this lemmatization

we should join all the words into meaning fully.

Q.No.

4) DBSCAN

~~Database~~

* It is represented in spherical shape.

* They are three types

(i) Core points

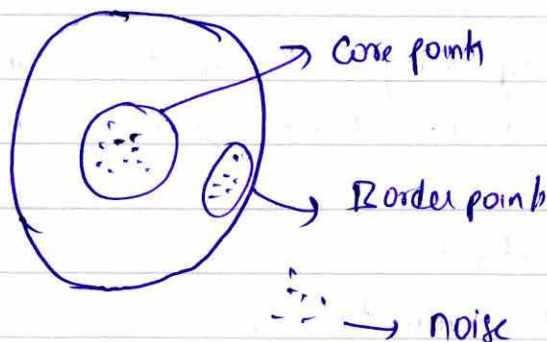
(ii) Border points

(iii) Noise

(i) Core points: The points which are in the radius of the spherical shape

(ii) Border points: It is present at the border of the spherical shape and it is neighbour

(iii) noise: The points which are outside of the spherical shape nor core point nor border points



Q.No.

Q.6) Tableau: It is software which is used to data visualization and it gives simple to understand the data in the graphs, bar charts etc.

* To learn tableau we should not need any programming skills

* Tableau is used by business purpose like the growth of the company, position of the company it will give like graph and simple to understand

* First we should visualize the data and then we use tableau to implement the graphs, charts etc.

* To create effective visualization first we should do data preprocessing methods to clean the data, it makes simple to analyze the data.

* It will adjust the screen for mobiles using tableau software

* Tableau software is efficient to use and we can learn easily by the data visualization

Q.No.

* For example

To check the status of the company it will give the average of working employees, growth of the company from year to year. Average income is get for the employees.

* It is the advanced software we use in data analysis

* It is used for visualisation the big data and efficient to use.

Q.No.

PART-A

✂ 1a) Here we use ETL in sand box is to used to evaluate the data, transmission the data.

- b) i) MATLAB
ii) TENSOR FLOW
iii) JUPYTER.

c) Text mining:

It is use to analyse the Text in simple way.

✂ The text data is to analyse and the large text data, in simple and ^{give} useful ~~in~~ information. removing noise words, using tokenization, part-of-speech etc.

d) Numpy it ~~can~~ will evaluate the mathematical operations. We use numpy in python because we don't need to write code from scratch, it is a multidimensional array.

✂ e) We use sample() in the data wrangling for

Q.No.

4)

g) joins it means we should join all the

h)

i) advantages

* It is more efficient to use ~~for~~ and we should not need any programming skills to use this

* We can adjust in the mobile screening while seeing the model









